# GlucoTwin

## An India-aware digital twin for proactive type 2 diabetes management

Team ZIZ, Chandigarh University

Happiest Health Digital Twin Challenge 2026



**Synthetic, open or anonymized data only**

**Decision support, not a medical device**

# Diabetes care is mostly reactive

### Seen every few months

Patients meet the doctor only every few months.

### Spikes noticed afterwards

A rise in glucose is usually seen after it happened.

### No continuous view

Between visits the doctor sees nothing continuous.

**Our goal: give the doctor a continuous view between visits, and warn about a likely excursion before it happens, not after.**

# What a digital twin is

A virtual copy of one patient, kept up to date with live data, that can monitor, predict and answer what-if questions.

### A normal prediction model

- ✗ Not personalized per patient
- ✗ Not continuously updated
- ✗ No what-if testing

### GlucoTwin digital twin

- ✓ Personalized per patient
- ✓ Continuously updated
- ✓ Supports what-if testing

# GlucoTwin at a glance



## Monitor

Live replay of the CGM stream



## Predict

Glucose at +30, +60 and +120 min with a q10 to q90 band



## Alert

Only events that pass a safety gate



## Explain

Top features behind each forecast



## Personalize

Small per-patient adapters



## What-if

Change a meal or activity and re-run



## Doctor dashboard

Five pages for the clinician

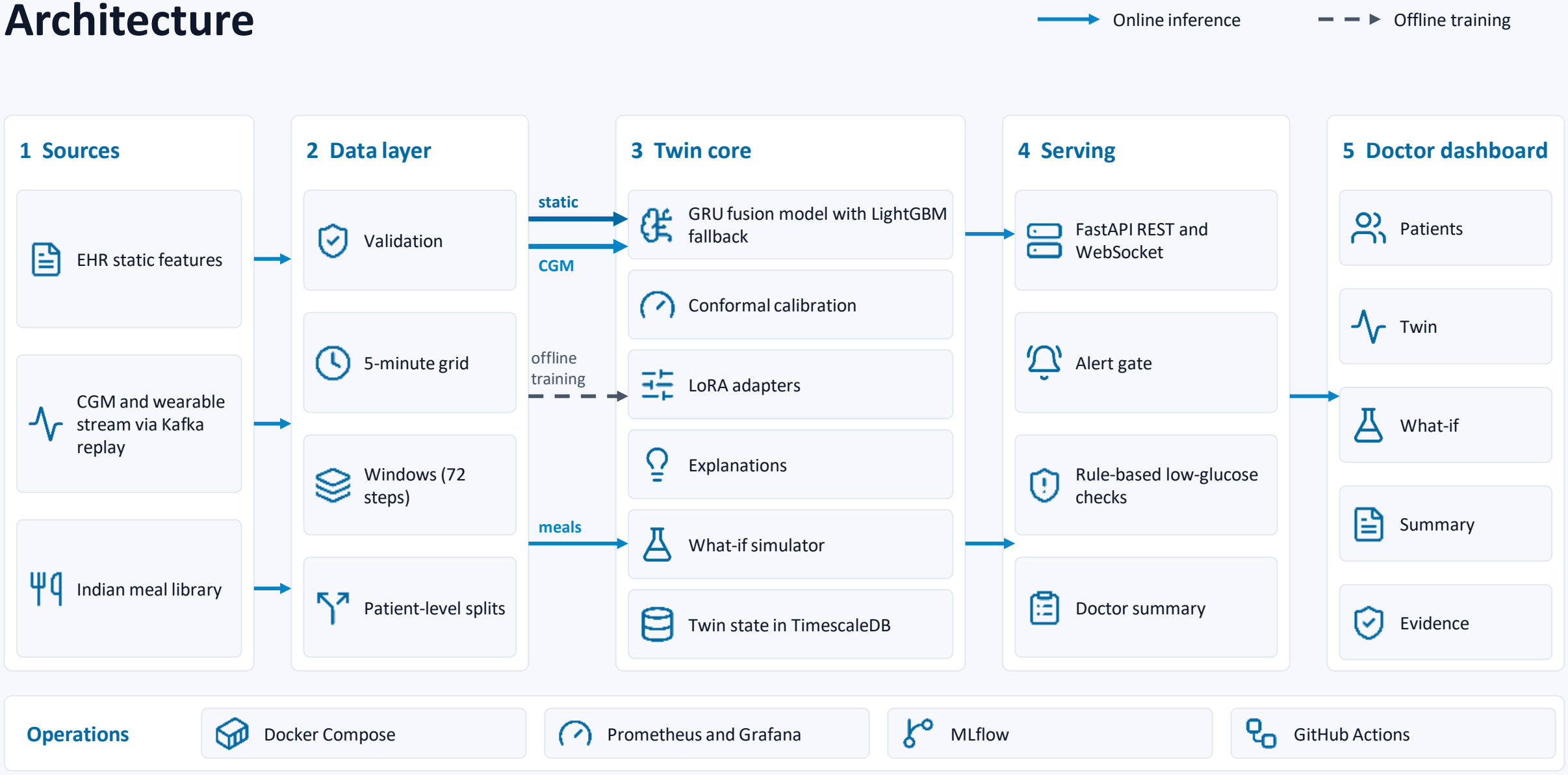
# Data sources, roles and licenses

| Source | Role | Size and notes | License |
|---|---|---|---|
| CGMacros (PhysioNet) | All training and evaluation | 45 participants (14 type 2 diabetes, 16 prediabetes, 15 healthy); 134,292 prediction windows | CC BY-NC-SA 4.0 |
| HUPA-UCM (Mendeley Data) | Optional pretraining only, never used for evaluation | Type 1, 25 patients | CC BY 4.0 |
| Synthetic AI-READI (ADVANCE Center) | Static patient table only | 10,518 synthetic static records | Terms not published |
| Indian meal library | What-if meals | 20 dishes; team estimates from IFCT 2017 raw ingredient values plus recipe assumptions; glycemic index values unverified | Team-made |

**No raw data redistributed**

**Record-to-wearable link is synthetic**

# Architecture



# How the model fuses two streams

### Static features

Health record data, turned into a static embedding

### CGM sequence

5-minute grid, 6-hour lookback (72 steps). Known-future inputs: hour of day, time since last meal, recent carbs.




### Outputs

- Forecasts at +30, +60 and +120 min
- Median plus q10 to q90 band
- Conformal calibration, 80% target

**80.2%**

of observed values fell inside the GRU interval at +30 min (80% target). Static and time-varying features are kept separate.

# Evaluation design: no patient leaks across folds

**5-fold leave-patients-out cross-validation**

**Stratified by diabetes status**

**Scalers fit on training patients only**

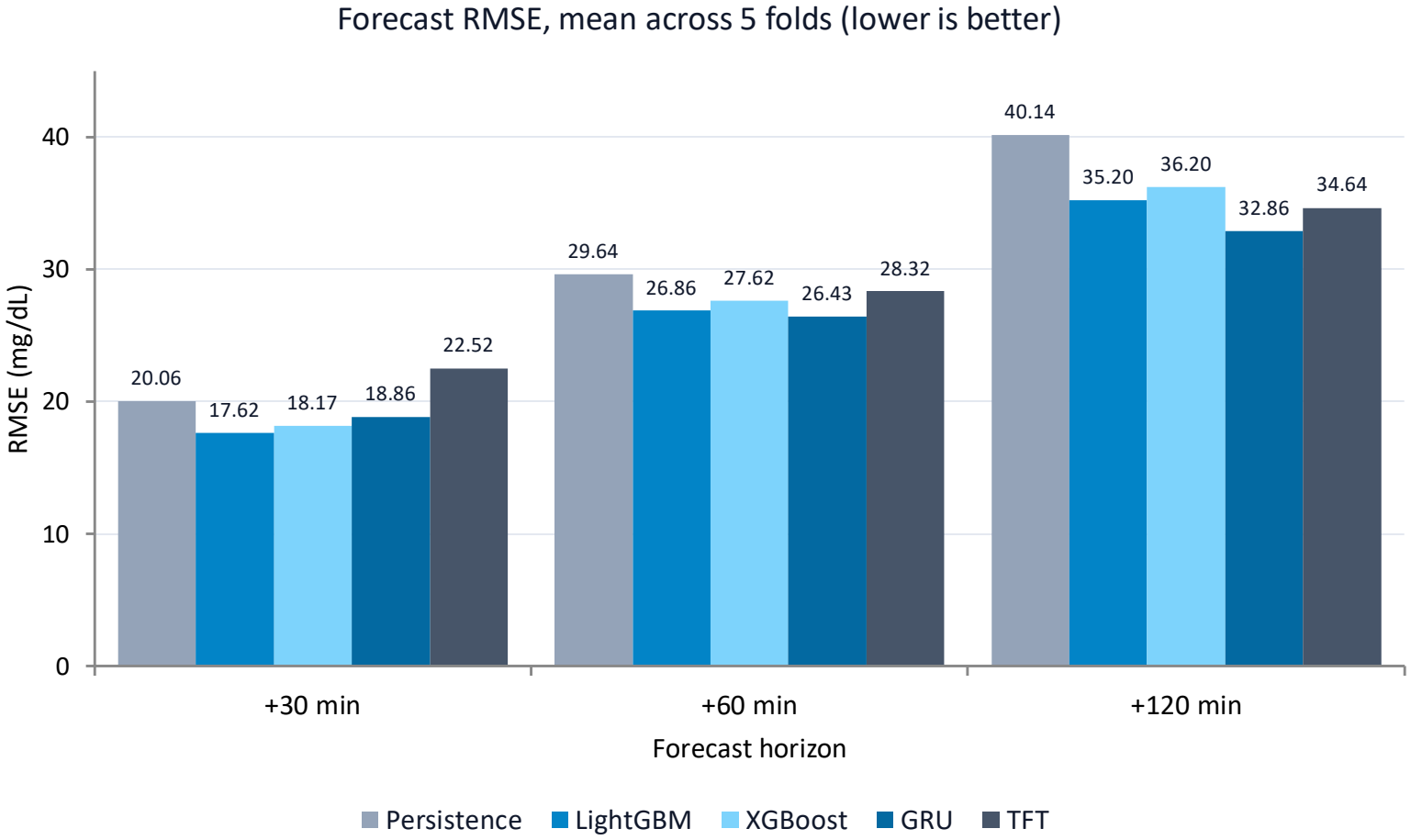
**No patient in more than one role**

## Events we predict

- **Hyperglycemia: above 180 mg/dL**
- **Severe: above 250 mg/dL**
- **Hypoglycemia: below 70 mg/dL**
- **Spike: rise of 40 mg/dL or more within 60 minutes**

134,292 prediction windows from 45 participants were used for all training and evaluation.

# Forecast error by horizon




**32.86**

mg/dL, GRU at +120 min, against 40.14 for persistence

A modest improvement over persistence. At +30 and +60 min the GRU, LightGBM and XGBoost are within one standard deviation of each other.

# Paired comparison with persistence

Mean reduction in per-patient RMSE over 45 patients, with 95% bootstrap interval

| Horizon | Model | Mean reduction (mg/dL) | 95% interval | Patients improved |
|---|---|---|---|---|
| +30 min | LightGBM | 2.39 | [1.92, 2.83] | 43 of 45 |
| +30 min | XGBoost | 1.99 | [1.35, 2.54] | 41 of 45 |
| +30 min | GRU | 1.41 | [0.83, 1.95] | 36 of 45 |
| +60 min | GRU | 3.21 | [2.09, 4.26] | 36 of 45 |
| +60 min | LightGBM | 2.85 | [1.85, 3.78] | 39 of 45 |
| +60 min | XGBoost | 2.36 | [1.16, 3.47] | 37 of 45 |

The intervals exclude zero, but the reductions are modest, between 1.41 and 3.21 mg/dL, and not every patient improves.

# Alerts only when a safety gate is passed

## The alert gate

- Held-out recall at least 40%
- Precision at least 30%
- Reliable: at least 10 events per fold
- At most 1.0 false alerts per patient-day

## Hyperglycemia at +30 min

**65 ± 10%**

recall

**61 ± 4%**

precision

**0.8 ± 0.2**

false alerts per patient-day

**18 min**

median lead time

## Five events pass

Hyperglycemia +30 min. Severe hyperglycemia +30 and +60 min. Spike +30 and +60 min. All other events are shown only as a risk estimate, never an alert.

## Low glucose: rule-based, not predicted

Predicted hypoglycemia did not pass (too few events). Checks on the observed reading: below 70, below 54, rapid fall.

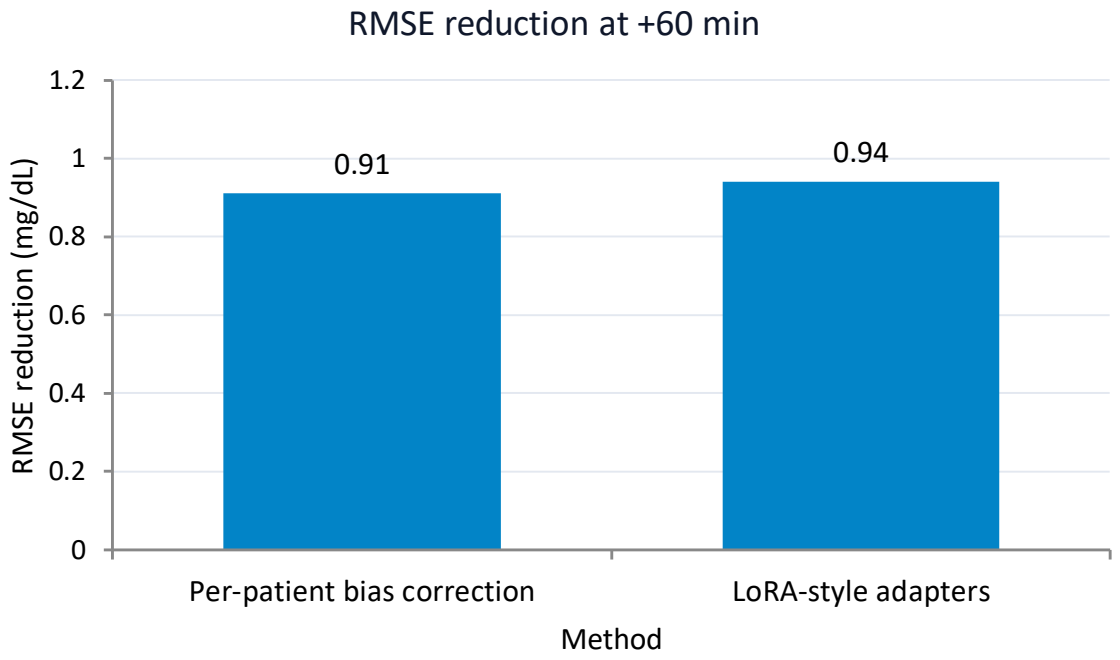
# Explanations and personalization

### Explanations

Permutation importance for the GRU, SHAP for LightGBM.

**39 of 45**

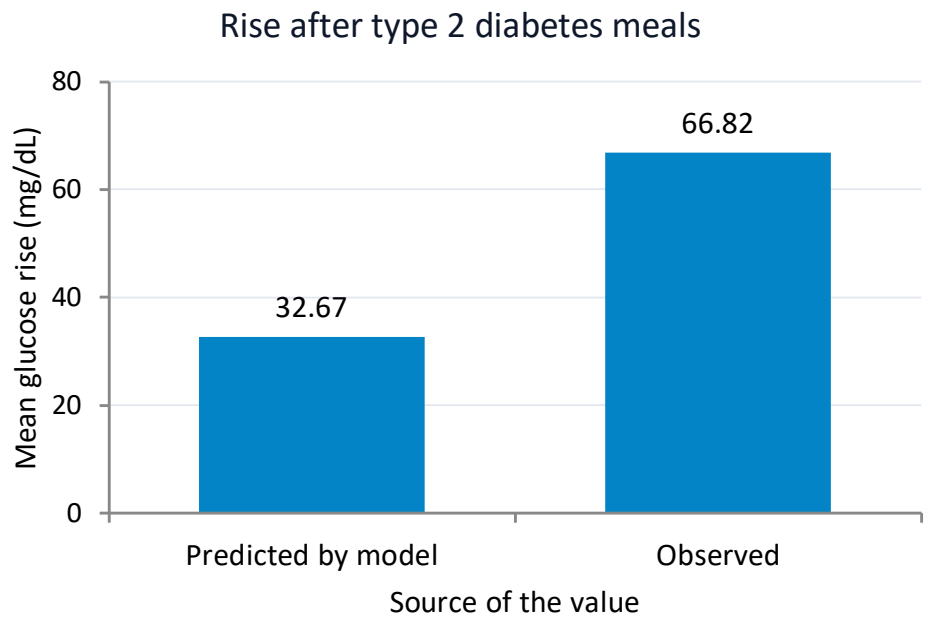
patients improved with per-patient adapters. MARD 12.82% to 11.90%. RMSE 24.14 to 23.20 mg/dL at +60 min.




A small effect. A simple per-patient bias correction gives almost the same RMSE gain (0.91 vs 0.94 mg/dL). Adapters are promoted only if held-out error improves.

# What-if simulator

Re-runs the model with a changed meal or activity. Default is the raw model output.




**98.4%**

of meals show a positive rise (direction is right)

**0.3994 ± 0.1102**

cross-fold Pearson r

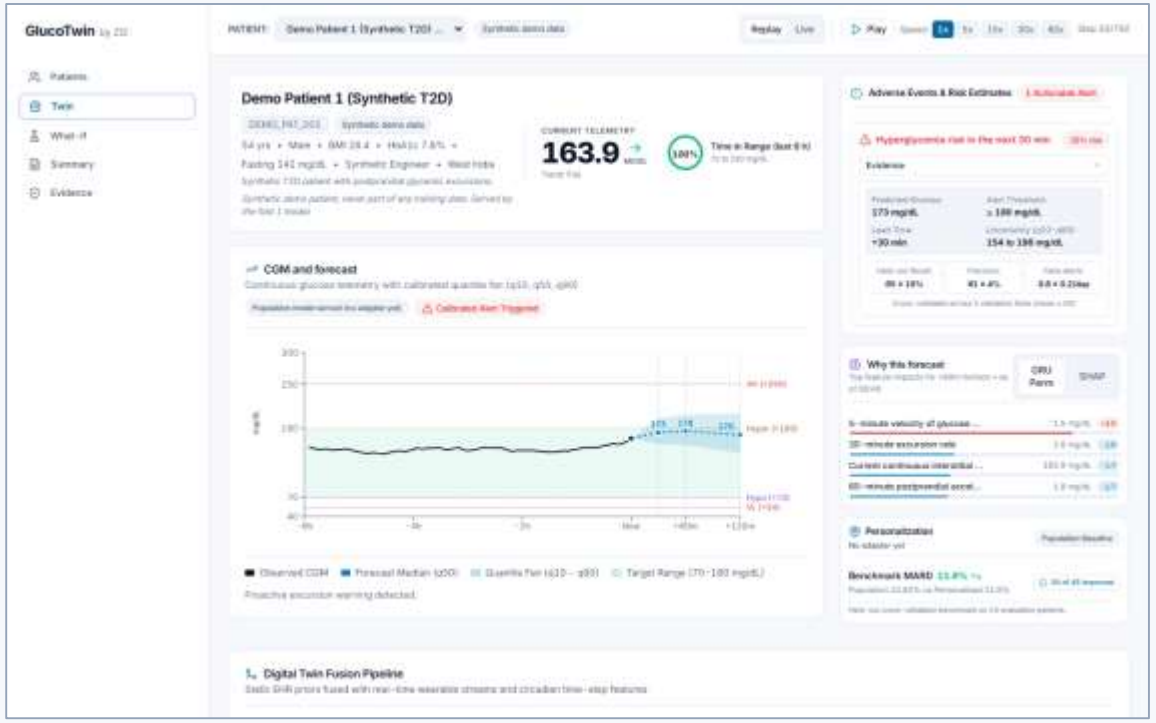
**72.5% to 100%**

monotonic in carbohydrate: raw model, then physiological projection (100% by construction)

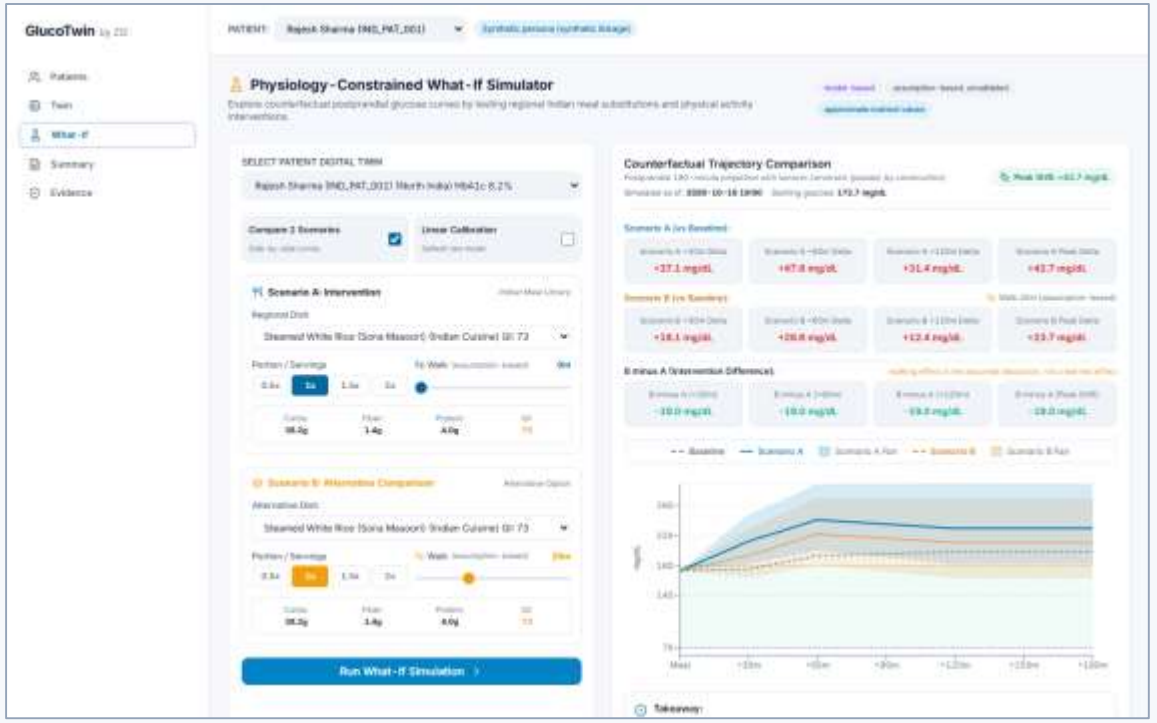
Meal rises are underestimated by about half. The walking effect is an assumption (0.38 mg/dL per net MET-minute, capped at 50 mg/dL), not validated.

model-based | assumption-based | approximate nutrient values

# Doctor dashboard



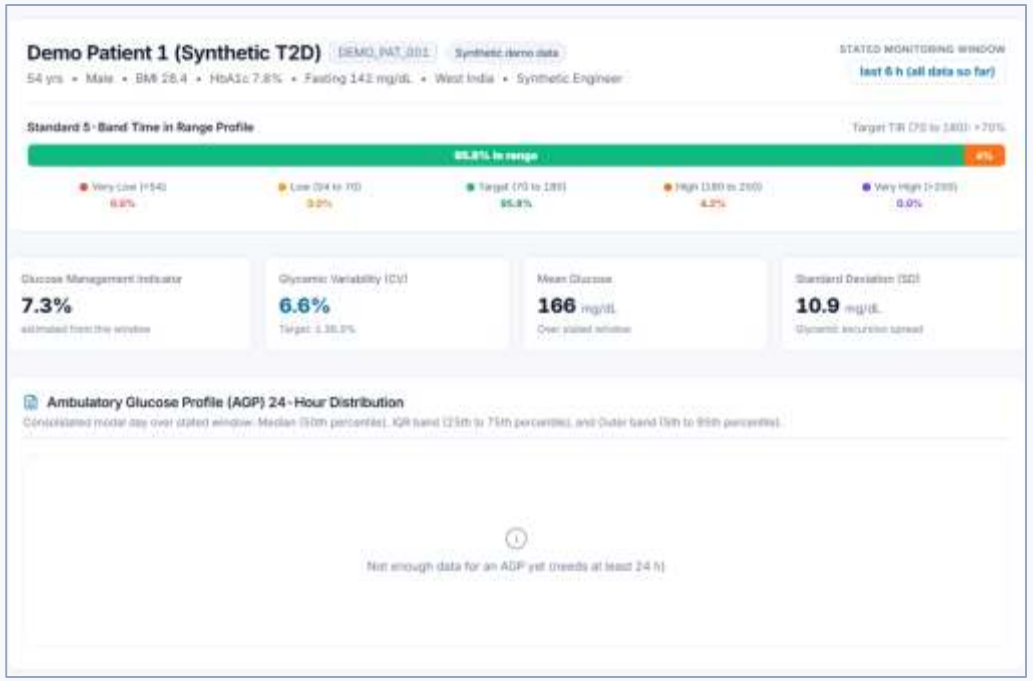

Twin page: replay, forecast fan, alert, explanation, personalization




What-if page: two scenarios against a baseline, with provenance badges

Five pages: Patients, Twin, What-if, Summary, Evidence. The demo uses 3 synthetic demo patients and 4 synthetic personas.

# Summary and Evidence pages




Summary page: time in range, GMI and CV. The AGP appears when at least 24 h of data exist.




Evidence page: every number is read from the saved reports.

The Evidence page lets a reviewer check the claims in this deck against the saved reports.

# Engineering and reproducibility

**125**

backend tests

**37**

frontend tests

**about 4.6 ms**

median inference latency

**about 34 to 44 ms**

median end-to-end stream latency, demo stack

### FastAPI

REST and WebSocket

### Kafka

Replay of the wearable stream

### PostgreSQL

with TimescaleDB

### Prometheus and Grafana

Monitoring

### MLflow

Experiment tracking

### Docker Compose

One command starts everything; backend image 2.37 GB

### GitHub Actions

4 jobs: lint and types, backend tests, frontend build, Docker build

### Licenses

Code MIT; models and reports CC BY-NC-SA 4.0

No raw data is included. Built with AI-assisted coding (Google Antigravity); the team specified, reviewed and tested everything.

# Limitations and safety

## Limitations

- Only 14 type 2 diabetes participants
- Record-to-wearable link is synthetic
- Hypoglycemia prediction is unreliable
- Meal rises underestimated by about half
- Meal nutrient values are estimates; glycemic index values unverified
- Activity effect is an assumption
- No prospective or clinical study

## Safety design

- Decision support, not a medical device
- No clinical validation, diagnosis or regulatory status is claimed
- Only five events can raise an alert
- Low-glucose checks are labeled rule-based, not predicted
- Every input shows a provenance badge

# Impact and roadmap

## What it offers

- A continuous view for the doctor between visits
- Predicted warnings for five event types instead of noticing spikes afterwards
- Indian meals for what-if questions, as estimates

## Roadmap

- Validate on larger Indian cohorts
- Device and FHIR integration
- Clinician feedback and usability testing
- Better hypoglycemia detection
- A prospective pilot
- Guideline-grounded explanations with citations (future work, not built)
- Photo-based meal logging (future work)

# Thank you

Team ZIZ, Chandigarh University

Happiest Health Digital Twin Challenge 2026

**Team Leader: Likhith Sai Seemakurthi**
**Member: Pritesh Kumar Pandey**

## GitHub repository

priteshkumarpandey/ZIZ_CHANDIGARH-UNIVERSITY

## Demo video

https://youtu.be/DC1_NouZk5E

# A1 Full RMSE table

Forecast RMSE in mg/dL, mean ± SD across 5 folds. Lowest in each column in blue.

| Model | +30 min | +60 min | +120 min |
|---|---|---|---|
| Persistence | 20.06 ± 0.89 | 29.64 ± 1.86 | 40.14 ± 3.15 |
| LightGBM | **17.62 ± 0.89** | 26.86 ± 2.77 | 35.20 ± 5.30 |
| XGBoost | 18.17 ± 0.96 | 27.62 ± 2.78 | 36.20 ± 5.16 |
| GRU (served primary) | 18.86 ± 1.01 | **26.43 ± 1.94** | **32.86 ± 2.63** |
| TFT (experiment) | 22.52 ± 1.87 | 28.32 ± 2.17 | 34.64 ± 3.07 |

Clarke error grid zones A+B at +30 min: LightGBM 99.39 ± 0.32 percent, GRU 99.34 ± 0.25 percent.

# A2 Alert operating table

Gate: recall at least 40%, precision at least 30%, reliable (at least 10 events per fold), at most 1.0 false alerts per patient-day.

| Event | Status | Operating point |
|---|---|---|
| Hyperglycemia +30 min | Alert | Recall 65 ± 10%, precision 61 ± 4%, 0.8 ± 0.2 false alerts per patient-day, median lead time 18 min |
| Severe hyperglycemia +30 min | Alert | See Evidence page |
| Severe hyperglycemia +60 min | Alert | See Evidence page |
| Spike +30 min | Alert | See Evidence page |
| Spike +60 min | Alert | See Evidence page |
| All other events | Risk estimate only | Not shown as alerts |
| Predicted hypoglycemia | Did not pass (too few events) | Rule-based checks on observed reading: below 70, below 54, rapid fall |

# A3 Data and license table

| Source or artifact | Role | License | Raw data redistributed |
|---|---|---|---|
| CGMacros (PhysioNet) | Training and evaluation | CC BY-NC-SA 4.0 | No |
| HUPA-UCM (Mendeley Data) | Optional pretraining only | CC BY 4.0 | No |
| Synthetic AI-READI (ADVANCE Center) | Static patient table only | Terms not published | No |
| Indian meal library | What-if meals (team estimates, IFCT 2017 based) | Team-made | No |
| Code | GlucoTwin software | MIT | Not applicable |
| Trained model files and reports | Served models and evidence | CC BY-NC-SA 4.0 (non-commercial, attribution, share-alike) | No raw data included |

# A4 Why not an LLM or RAG?

We did not build one. The task is numeric forecasting of a glucose time series with uncertainty, and our models can be tested on held-out patients. Our explanations come from permutation importance and SHAP, which point to model inputs. Guideline-grounded explanations with citations are future work and are not built.

# A5 Why a GRU and not a transformer?

We tried a Temporal Fusion Transformer as an experiment. At +30 and +60 min it had the highest RMSE of the learned models (22.52 and 28.32 mg/dL), and at +120 min the GRU was lower (32.86 against 34.64). The GRU is lowest at +60 and +120 min and fuses the static embedding with the CGM sequence. LightGBM is lowest at +30 min, within one standard deviation, so it is the fallback.

# A6 How do you know it generalizes?

We only know it partly. Evaluation uses 5-fold leave-patients-out cross-validation, stratified by diabetes status, with scalers fit on training patients only, and paired comparisons with bootstrap intervals over 45 patients. But there are only 45 participants, 14 with type 2 diabetes, the record-to-wearable link is synthetic, and there is no prospective or clinical study.

# A7 Is hypoglycemia handled?

Not by prediction. Predicted hypoglycemia did not pass the alert gate because there were too few events, so it is shown only as a risk estimate. Low glucose is covered by rule-based checks on the observed reading (below 70, below 54, rapid fall), labeled rule-based, not predicted. Better hypoglycemia detection is on the roadmap.

# A8 What are the main limits?

Only 14 type 2 diabetes participants; a synthetic link between record and wearable; unreliable hypoglycemia prediction; meal rises underestimated by about half; estimated meal nutrient values and unverified glycemic index values; an activity effect that is an assumption; and no prospective or clinical study.